Proofs and corrections of OEIS conjectures

29 papers in six families and a group of other results

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This collection settles conjectures recorded in 29 entries of the On-Line Encyclopedia of Integer Sequences (OEIS). Most of the sequences are coefficients of formal power series defined by functional equations, and most of the conjectures concern their residues modulo small integers. In 24 entries the conjectures treated are proved, and in A392210 an observation recorded in the entry is proved. In four entries (A380680, A396798, A396843 and A396844) a conjectured statement is false; the papers give counterexamples and prove the correct statements. Each paper is self-contained, and its main theorems are formalized in Lean 4 with Mathlib; the formal proofs use only Lean's standard axioms. All statements were also checked numerically against the published OEIS data.

The results were obtained in an AI research effort directed by Dev D. Goyal: ChatGPT (OpenAI, GPT-6 with Pro reasoning), working autonomously from Dev D. Goyal's prompts, found the proofs and wrote the Lean formalization, and Claude Opus 5.5, used with Claude Code, checked the work independently and wrote the papers in their present form. Dev D. Goyal has reviewed the proofs; this review is not an independent expert validation. The papers have not been peer reviewed.

1. The family $A_m(xA_m - mxA_m^2) = x^2$

Parity, residues modulo m^2 and binary-digit divisibility for the unique normalized solution of $A_m(xA_m - mxA_m^2) = x^2$, with corrected residue classifications for the two entries.

A396843. Odd coefficients, remainders modulo 9, and binary digits for the series $A(xA - 3xA^2) = x^2$

Every coefficient is odd; the exact residues modulo 9 (correcting the entry's second conjecture); $3^{s_2(n-1)}$ divides $a(n)$, where s_2 is the binary digit sum. *(proved; one statement corrected.)*

Paper · OEIS A396843 · Lean

A396844. Remainders modulo 16 for the series $A(xA - 4xA^2) = x^2$

The exact residues modulo 16; the conjecture fails first at $n = 17$. *(conjecture disproved; corrected statement proved.)*

Paper · OEIS A396844 · Lean

2. Consecutive iterates $F = x + F^{\circ r}\, F^{\circ(r+1)}$

For every integer k, the k-th iterate of F_r modulo $2r$ is $x/(1-kx)$ when r is odd, with an explicit correction term when r is even.

A396099. The iterates of a self-referential power series modulo 4

All four conjectured congruences modulo 4 for the series and its iterates. *(proved.)*

Paper · OEIS A396099 · Lean

A396798. The iterates of a self-referential power series modulo 8

Seven of the eight conjectured congruences modulo 8; the remaining one holds from $n = 2$, not from $n = 1$ as stated. *(proved; one statement corrected.)*

Paper · OEIS A396798 · Lean

A396807. The iterates of a self-referential power series modulo 10

For every integer k, the coefficients of the k-th iterate satisfy $[x^n]\, A^{\circ k} \equiv k^{n-1} \pmod{10}$. *(proved.)*

Paper · OEIS A396807 · Lean

3. Scaled inverses $A(x - x\,A(qx)/q) = x$

Every coefficient is 1 modulo $q+1$; for even q, a coefficient is odd exactly when its index is a power of 2.

A393856. Congruences for the series A with $A(x - x\,A(4x)/4) = x$
$a(n) \equiv 1 \pmod 5$, and $a(n)$ is odd exactly when n is a power of 2. *(proved.)*
Paper · OEIS A393856 · Lean

A393857. Congruences for the series A with $A(x - x\,A(5x)/5) = x$
$a(n) \equiv 1 \pmod 6$. *(proved.)*
Paper · OEIS A393857 · Lean

A393858. Congruences for the series A with $A(x - x\,A(6x)/6) = x$
$a(n) \equiv 1 \pmod 7$, and $a(n)$ is odd exactly when n is a power of 2. *(proved.)*
Paper · OEIS A393858 · Lean

A393859. Congruences for the series A with $A(x - x\,A(7x)/7) = x$
$a(n) \equiv 1 \pmod 8$. *(proved.)*
Paper · OEIS A393859 · Lean

4. Theta series and theta-type equations

Series built from the Jacobi theta series θ_3 or defined by theta-type sums, whose coefficients follow simple patterns modulo 4 or 16.

A378580. Remainders modulo 4 for the series with $A(x/A(x)) = \theta_3(x)$
$A(x/A(x)) = \theta_3$: for $n > 0$, $a(n) \equiv 2 \pmod 4$ if n is a square, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A378580 · Lean

A378581. Remainders modulo 4 for the series with $A(x\,A(x)) = \theta_3(x)$
$A(x\,A(x)) = \theta_3$: the same square pattern. *(proved.)*
Paper · OEIS A378581 · Lean

A380060. Square indices and remainders modulo 4 in a theta-type series
For $n > 0$, $a(n) \equiv 2 \pmod 4$ if n is a square, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A380060 · Lean

A380680. Twice the squares, not the squares: remainders modulo 4 in a theta-type series
The square conjecture is false ($a(1) = 4$, $a(2) = 6$); in fact $a(n) \equiv 2 \pmod 4$ exactly when $n = 2k^2$. *(conjecture disproved; corrected statement proved.)*
Paper · OEIS A380680 · Lean

A355872. Residues modulo 16 for a series defined by a bilateral theta-type equation
$a(n) \equiv 2(-1)^{n-1} \pmod{16}$ for all $n \geq 1$, which implies the three conjectured congruences modulo 4 and 8. *(proved.)*
Paper · OEIS A355872 · Lean

5. Catalan-type equations

Equations whose solutions reduce, modulo 2, 4 or 8, to expressions in the series $\sum_{n\geq 1} C_{n-1}x^n$ of Catalan numbers, whose n-th coefficient is odd exactly when n is a power of 2.

A393172. The Catalan numbers inside the equation $A(x - x^2) = x + A(x)^2$
For $n > 1$, $a(n) \equiv 2 \pmod 4$ if $n = 2^k$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A393172 · Lean

A393171. Remainders modulo 8 of a series built from the Catalan numbers
For $n > 2$, $a(n) \equiv 4 \pmod 8$ if $n = 2^k$ or $2^k + 1$, and $8 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A393171 · Lean

A389536. Remainders modulo 4 for the equation $A(x^2 + 2x^3) = x\,A(x)$
For $n > 1$, $a(n) \equiv 2 \pmod 4$ if $n = 2^k + 1$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A389536 · Lean

A397588. Odd coefficients at powers of two for $A = x + (xA^2)'$
$a(n)$ is odd exactly when n is a power of 2, and $3 \mid a(n)$ for $n > 1$. *(proved.)*
Paper · OEIS A397588 · Lean

A195194. Parity of an infinitely nested series reversion
The infinitely nested reversion is well defined, and $a(n)$ is odd exactly when n is a power of 2. *(proved.)*
Paper · OEIS A195194 · Lean

6. Equations with an exact halving step

Equations whose analysis divides by 2: in A389474, A389475 and A389542 the coefficient recursion divides by 2, and the division is shown to be exact; in A389537 the series $(A - x - x^2)/2$ is shown to have integer coefficients and is studied modulo 2.

A389537. Congruences modulo 4 for the series with $A(A(x) - x) = A(x)^2$
For $n > 2$, $a(n) \equiv 2 \pmod 4$ if $n = 3 \cdot 2^k$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A389537 · Lean

A389542. Congruences modulo 4 for the series with $A\big(A(x)^2 - x^2\big) = 4A(x)^3$
For $n > 1$, $a(n) \equiv 2 \pmod 4$ if $n = 2^k + 1$, and $4 \mid a(n)$ otherwise. *(proved.)*
Paper · OEIS A389542 · Lean

A389474. A power series whose square is $x^2 + 2x^3$ modulo 9
$A^2 \equiv x^2 + 2x^3 \pmod 9$, and $A^{3k} \equiv A(x^3)^k \pmod 3$. *(proved.)*
Paper · OEIS A389474 · Lean

A389475. Two congruences modulo 3 for the series with $A(3x^2) = 2x^2 + A^2 - 2A^3$
$A^3 \equiv A(x^3)$ and $A^2 \equiv x^2 + 2A(x^3) \pmod 3$. *(proved.)*
Paper · OEIS A389475 · Lean

7. Other results

Further entries, each with its own method.

A396093. When are the coefficients of $B(B(B(x)))$ odd?
$a(n)$ is odd exactly when $n \equiv 1, 3, 7, 9 \pmod{10}$. *(proved.)*
Paper · OEIS A396093 · Lean

A392210. The odd-indexed terms of a nested square are even after the first term
$a(2n-1)$ is even for $n > 1$ (an elementary observation recorded in the entry). *(proved.)*
Paper · OEIS A392210 · Lean

A396102. Every coefficient is 1 modulo 3: the series with $A(A(A(x))) = (1 + x)\,A(A(x))$
$a(n) \equiv 1 \pmod 3$, together with a theorem for the whole family $F^{\circ r} = (1 + x)F^{\circ(r-1)}$. *(proved.)*
Paper · OEIS A396102 · Lean

A380056. The numbers with exponential generating function $(e^x - 1)/\cos 2x$, modulo 4 and 5
$a(n) \equiv 1 \pmod 4$, and $a(n) \equiv 2 \cdot 2^n + 3 \cdot 4^n \pmod 5$, so $5 \mid a(n)$ exactly when $4 \mid n$. *(proved.)*
Paper · OEIS A380056 · Lean

A397346. Residues modulo 8 for a series defined by a self-referential exponential

From $n = 2$ on, $a(n)$ mod 8 repeats $4, 2, 0, 2$; more precisely, $a(n)/n \equiv 6 \pmod 8$ if $n \equiv 3 \pmod 4$ and $a(n)/n \equiv 2 \pmod 8$ otherwise. *(proved.)*

Paper · OEIS A397346 · Lean

A394576. A greedy sequence that avoids sums of three consecutive terms

$\lfloor (4n-3)/3 \rfloor \le a(n) \le \lfloor (4n-1)/3 \rfloor$ for all $n \ge 1$; in particular $a(3k) = 4k - 1$. *(proved.)*

Paper · OEIS A394576 · Lean